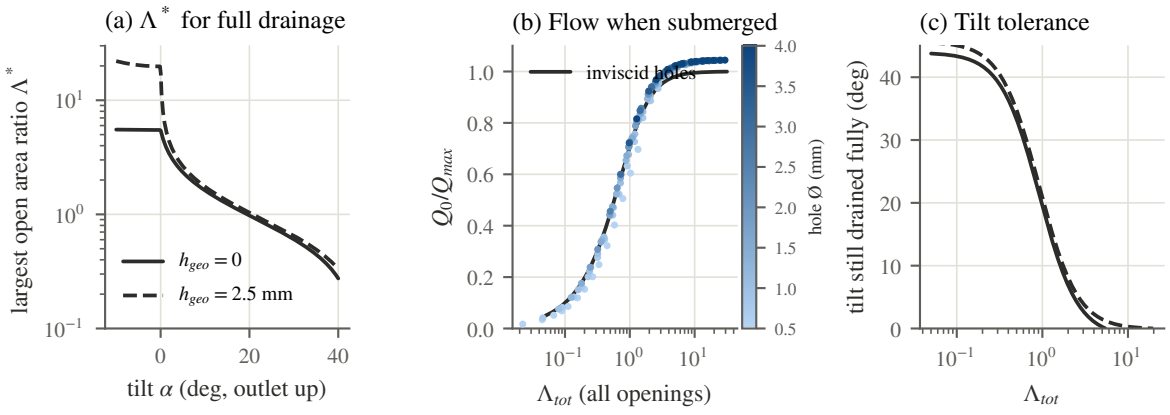
(a) Λ^* for full drainage
largest open area ratio Λ^*
10^1
10^0
10^{-1}
$h_{geo} = 0$
$h_{geo} = 2.5$ mm
0
20
40
tilt α (deg, outlet up)
(b) Flow when submerged
inviscid holes
Q_0/Q_{max}
1.0
0.8
0.6
0.4
0.2
0.0
10^{-1}
10^0
10^1
Λ_{tot} (all openings)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
hole Ø (mm)
(c) Tilt tolerance
tilt still drained fully (deg)
40
30
20
10
0
10^{-1}
10^0
10^1
Λ_{tot}